

Unit 1 Comprehensive Review

Directions: No calculators are allowed. Show all work. Reduce all fractions.

A. Multiplication

1. Evaluate.

a. $(7)(8) =$

b. $(6)(9) =$

c. $(12)(4) =$

d. $(11)(7) =$

e. $(8)(8) =$

f. $(9)(3) =$

g. $(6)(6) =$

h. $(12)(7) =$

i. $(4)(9) =$

j. $(8)(5) =$

k. $(11)(6) =$

l. $(7)(7) =$

B. Positive and Negative Integers, Square Roots, Exponents, Order of Operations

1. Evaluate.

a. $42 + (-17) =$

b. $-13 + 6 =$

c. $51 - (-24) =$

d. $-28 - 17 =$

e. $-15 + (-9) =$

f. $-5 - (-12) =$

g. $6 + (4)(5) =$

h. $(7 - 2^2)(-3) =$

i. $(3 + 5)^2 - 2^3 =$

j. $\sqrt{25} + 11 =$

k. $3^4 =$

l. $2^8 =$

m. $\sqrt{144} =$

n. $\frac{(2 + 3)^2 - (-1 + 11)^2}{5} =$

o. $18 - (3 + 2)^2 =$

p. $\sqrt{81} - (-4) + (2)(3) =$

C. Fractions/Decimals, Percent

1. Evaluate. Reduce all fractions.

a. $(0.6)(0.07) =$

b. $\frac{8.4}{0.7} =$

c. $7.35 + 2.4 =$

d. $14.8 - 6.25 =$

e. $(1.2)(0.7) =$

f. $(\frac{2}{3})(\frac{3}{4}) =$

g. $\frac{2}{3} + \frac{1}{4} =$

h. $\frac{\frac{3}{5}}{\frac{9}{10}} =$

i. $(\frac{2}{3})^3 =$

j. $\left(-\frac{2}{3}\right)^4 =$

k. $\frac{7}{12} - \frac{1}{4} =$

l. $\frac{5}{6} + \frac{1}{3} =$

m. $\frac{42}{56} =$

n. $\sqrt{\left(\frac{9}{4}\right)(16)} =$

o. $\frac{3}{5} + \frac{1}{2} - \frac{1}{4} =$

p. $\frac{\sqrt{8^2 + (4)(9)}}{\sqrt{100}} =$

2. Find each amount:

a. What is one eighth of 56?

b. What is three fifths of 40?

c. What is seven tenths of 90?

3. Find each percent:

a. What is 12% of 250?

b. What is 35% of 80?

c. What is 125% of 48?

4. The value of a rare card is now 180% of its original value of \$250. What is it worth now?

5. A \$250 tablet is on sale for 20% off. After the discount, 8% sales tax is charged on the new price. What is the final cost?

6. A pair of shoes costs \$160 and is on sale for 25% off. After the discount, 10% sales tax is charged on the new price. What is the final cost?

7. A school event has 400 people attend this year. Attendance increases by 25% the next year, then increases by 10% of that new total the following year. How many people attend after both increases?

8. A bicycle costs \$300 and is marked 20% off. At checkout, you receive an additional 25% off the sale price. What is the final price?

9. A store buys an item for \$80 and marks the price up by 50%. Later, the store puts that new price on sale for 25% off. What is the sale price?

10. A \$200 jacket is on sale for 25% off. After the discount, 8% sales tax is charged on the new sale price. What is the final cost?

11. A tournament has 200 registered players. Registration increases by 30% the next year, but then 10% of that new total does not show up. How many players actually attend?

D. Types of Numbers (Rational, Irrational, Integer, Whole, Natural)

1. Simplify each number or expression first. Then classify it in as many ways as possible using the five categories below:

Integer

Irrational

Natural

Rational

Whole

a. 19

h. $6.272727\ldots$

b. -4.6

i. $\sqrt{13}$

c. π

j. $\frac{72}{18}$

d. $\frac{11}{4}$

k. $\frac{42}{56}$

e. 0

l. $-\sqrt{64}$

f. -7

m. $\sqrt{121}$

g. $5.101001000100001\ldots$

n. 8.0